

How prior experience with pitch accents shapes the perception of word and sentence stress

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ABSTRACT

Listeners perceive high or rising pitch as stressed – at the word and sentence level (high-pitch bias). Since stressed syllables can also be low-pitched, this bias may lead to misinterpretations of word and sentence stress and thus slow down speech comprehension. We investigate the effect of immediate exposure with high- vs. low-pitched stressed syllables on the identification of word and sentence stress. Participants were exposed to utterances containing only high- vs. low-pitched stressed syllables. In experimental trials, they then heard either trisyllabic words with word stress on the second syllable (Experiment 1) or three-word-sentences with sentence stress on the second word (Experiment 2) and indicated the position of word/sentence stress. Stimuli were presented in three intonation conditions (high-pitched first, second, or third syllable/word). Both experiments endorsed the high-pitch bias, with an increase after high-pitch exposure. Our results suggest a speaker-independent re-weighting of acoustic cues to stress, which is driven by experience.

KEYWORDS

Experience; processing mechanism; stress perception; sentence stress; word stress; intonation; German

1. Introduction

In spoken language, as in other perceptual domains, some items “stand out”, i.e. are more salient or noticeable than others. Similar to a visual-search task in which a red object amongst a group of green ones is easily detected (Theeuwes, 1995; Treisman & Gelade, 1980), the stressed syllable within a spoken word perceptually stands out (word stress); likewise, accented words stand out from a spoken utterance (sentence stress). The placement of stress in words and sentences has numerous consequences for linguistic processing, ranging from lexical activation to the interpretation of the intended information structure. The notion of “stress” – both on the word and sentence level – is syntagmatic. In this sense, “stress” is a categorical concept: on the word level, a syllable can be either stressed¹ or unstressed and is processed as such (Connell et al., 2018; Cooper et al., 2002; Cutler & Norris, 1988; Domahs et al., 2008; Friedrich et al., 2004; Reinisch et al., 2010); this is also true on the phrase level where words can be accented or not (cf. Gussenhoven, 2011), henceforth referred to as sentence stress. We focus on the perception of prominence as a syntagmatic, categorical concept on two linguistic levels: the perception of word stress on the level of the word and the perception of sentence stress on the

level of the phrase, using a similar paradigm. Lexical and phrasal prominence are related because pitch movements, which are a defining characteristics of prominence-lending pitch accents in intonation languages, are associated with lexically stressed syllables (Ladd, 2008). Pitch movements may be aligned or misaligned with the stressed syllable as a function of different pitch accent types (Ladd, 2008), which renders stressed syllables either high- or low-pitched.

Research on word-stress perception has established that high pitch is a strong perceptual cue to the position of lexical stress (see early studies by Fry, 1958 on English; Isačenko & Schädlich, 1966 on German). These early studies have often been criticised for conflating phrase-level accentuation and word-level stress because the stressed syllable was always accented (e.g. Roettger & Gordon, 2017 for discussion).² However, in natural speech, stressed syllables are quite often accented and hence accompanied by an f₀-movement: In German, for instance, 44% of the words are accented, i.e. nearly every second word (Antje Schweitzer (p.c.); for similar numbers on English see Dainora, 2006). It seems that listeners find it difficult to dissociate the two levels of prominence: lexical stress and phrase-level accentuation, i.e. f₀-movement: A number of studies recently

reported that the perception of word-level stress is influenced by phrase-level intonation, i.e. whether stressed syllables are high- or low-pitched (Dilley & Heffner, 2013; Friedrich et al., 2001; Schwab & Dellwo, 2017; Zahner et al., 2019); unstressed syllables were even misinterpreted as being stressed when they were high-pitched (Zahner et al., 2019). On the level of the phrase, similar findings have been reported for perceived prominence of rising or high pitch in spoken utterances (Baumann & Röhr, 2015; Baumann & Winter, 2018; Niebuhr & Winkler, 2017; Wagner et al., 2016) with a focus on degrees of perceived prominence (e.g. Baumann & Winter, 2018; Cole et al., 2010; Terken, 1991). Given the link between the word and phrase level, it is not surprising that pitch affects both, the perception of lexical stress and the perceived prominence at the utterance-level. We will henceforth use the term “high-pitch bias” as a descriptive term for this finding. While the “high-pitch bias” is well documented on both linguistic levels (word level and phrase level), we know little about whether the high-pitch bias is comparable at the word and the phrasal level. Hence, our first aim is to directly compare prominence perception on the two linguistic levels, using the same paradigm. Our second, and more important aim is to test whether the high-pitch bias (observed on both levels) is caused by an inherent salience of high pitch or arises as a result of experience.

The main aim of this paper is thus to investigate the source of the high-pitch bias, both for the identification of word stress and sentence stress. One potential source of the high-pitch bias is the frequency of occurrence of high-pitched stressed syllables in the input (henceforth referred to as experience, cf. Behrens & Pfänder, 2016 for a summary of frequency effects in other linguistic domains). The idea is the following: If stressed syllables are predominantly high-pitched, then listeners are in turn more likely to interpret a high-pitched syllable as stressed, which reinforces the link between high pitch and stress. Such a mechanism might influence the weighting of cues towards a high-pitch bias in the perception of both word and sentence stress. Furthermore, the re-weighting of cues may eventually lead to misinterpretations of a words’ metrical structure and prominence at the sentence level and thus slow down speech comprehension. In the present study, we directly test whether the exposure with high- vs. low-pitched stressed syllables affects the weighting of the pitch peak as a cue to prominence. Different exposure phases hence serve as a methodological tool to tap into the role of experience. Before turning to the details of the present study (1.3), we will elaborate on the link between word stress and phrase-level intonation

and the processing thereof (1.1), and outline the role of experience in speech processing (1.2).

1.1. The link between word level and sentence level prominence

Stressed syllables play a role for speech processing in so-called stress languages, such as English, Dutch, German, Spanish and Italian (e.g. Cooper et al., 2002; Cutler & Norris, 1988; Donselaar et al., 2005; Friedrich et al., 2004; Reinisch et al., 2010; Soto-Faraco et al., 2001; Sulpizio & McQueen, 2012; Zahner et al., 2019). In these languages, each content word contains one syllable with primary stress, which functions as the head of the word in a phonological sense (Hyman, 1977), e.g. the second syllable of the word “banana”. Acoustically, stressed syllables have longer duration, higher intensity and more peripheral vowel quality (Gordon & Roettger, 2017, for an overview). In many languages, stressed syllables additionally serve as anchors for phrase-level pitch accents, which are signalled by f_0 -movements, perceived as pitch changes (Ladd, 2008). There are a range of different pitch accent types, which signal differences in information status, communicative intent, speaker biases, among others (see Ladd, 2008 for an overview). For instance, if we answer the question “What did you eat?” in a neutral way with the word “A banana!”, the stressed syllable “na” will most likely be high-pitched and preceded and followed by a low-pitched syllable, respectively. On the other hand, if we question that the dog ate the fruit, we will most likely say something like “A banana? Really?” and produce the word “banana” with a low-pitched stressed syllable followed by high pitch on the final syllable. Hence, the peak of an f_0 movement may be realized directly on the stressed syllable (“banana!”) or on the syllable following the stressed one (“banana?”); in other contexts, it might also be realized on the syllable preceding the stressed one (see Figure 1C for an overview of the three contours used in the experiments).

In intonational phonology, pitch accents are described as a sequence of high (H) and low (L) tones, which receive diacritics to indicate their association with the stressed syllable (H* indicates an accent with high pitch on the stressed syllable) and with boundary tones (H% indicates high pitch at the end of the phrase). Henceforth, we use the term “medial-peak accent” to refer to an H* accent, in which the pitch peak is aligned with the stressed syllable, the term “early-peak accent” to refer to accents in which the pitch peak precedes the stressed syllable, rendering the stressed syllable itself low-pitched (H + L*), and

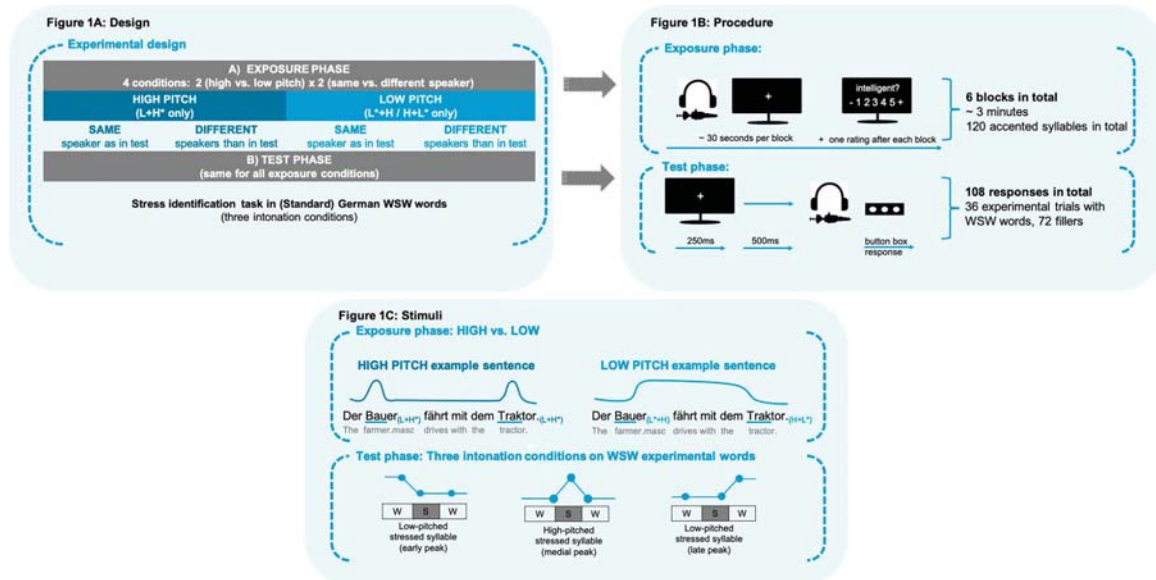


Figure 1. Overview of experimental design (A), procedure of exposure and test phase (B), and examples of stimuli in both phases (C) of Experiment 1.

finally the term “late-peak accent” to refer to accents in which the pitch peak follows the stressed syllable, also rendering the stressed syllable low-pitched (L^*+H). Consequently, while the position of word stress is fixed (in our design, the second syllable, e.g. “banana”), the pitch peak is an unreliable cue for the position of the stressed syllable. For early- and late-peak accents the position of the pitch peak might be misleading, in fact, pointing to the first or last syllable, respectively. Indeed, recent studies have documented such effects of phrase-level intonation on stress (Dilley & Heffner, 2013; Friedrich et al., 2001; Schwab & Dellwo, 2017; Zahner et al., 2019). For example, Dilley and Heffner (2013) showed that American English listeners more often indicated primary stress on syllables that were high-pitched in English polysyllabic words presented in different intonation contours. Similarly, Zahner et al. (2019), Experiment 1, demonstrated that German listeners gave more accurate stress judgements on German trisyllabic words when the stressed syllable was high-pitched (H^* -accent) than when the pitch peak preceded or followed the stressed syllable. Intonation has further been shown to directly affect lexical activation in a visual-world eye-tracking experiment (Zahner et al., 2019, Experiment 2): German listeners erroneously activated SWW (strong-weak-weak) words (e.g. *Libero* [ˈli:bero] “sweeper”) when the first syllable in auditorily presented – and segmentally overlapping – WSW words (e.g. *Libelle* [liˈbɛlə] “dragonfly”) was high-pitched (as in early-peak accents, $H+L^*$).

The same reasoning applies for pitch-accented words within a sentence where the perception of sentence

stress (i.e. most prominent word) may be guided by the position of the pitch peak: In our design, we concentrate on *wh*-questions (e.g. “Who drinks milk?”) with the focus on the verb – signalled by prosodic prominence –, which are compatible with situations that question the lexical semantics of the verb (e.g. “drinking” vs. “buying”) or its polarity (“drinking” vs. “not drinking”). In these sentences, relying on the pitch peak to identify the position of the stressed word (sentence stress) might point to the verb in case of a medial-peak accent. An early-peak accent on the verb, in turn, could be misinterpreted as an accent on the *wh*-word (as in an echo-question), and a late-peak accent on the verb could be misinterpreted as an accent on the final noun (to question the object, see Kutscheid et al., 2021). As mentioned above, the present study directly compares the effect of pitch accent type on stress perception on word and sentence level – using the same paradigm.

We predict that stress judgements on the sentence level might be more strongly affected by intonation than stress judgements on the word level: In German, word stress is lexically determined while sentence stress can, in theory, fall on any word in the sentence. Specifically, stress judgements on the word level may be affected by top-down information, i.e. lexical information or stress-rules (e.g. Levelt et al., 1999; Protopapas et al., 2016; Wagner, 2005) and/or by bottom-up information, i.e. acoustic stress-cues in the signal (e.g. Fry, 1958; Kohler, 2009, 2012; Sluijter et al., 1997). Word stress is a concept that is taught in schools: some orthographic rules in German rely on the position of word stress (e.g. consonant doubling) and stress plays a role

in poetry (verse and metre). The functional load of word stress in German is rather small; there are only a small number of pairs that only differ in word stress (e.g. *modern* “modern” (WS) vs. “to rut” (SW), *umfahren* “to bypass” (WSW) vs. “to knock over with a vehicle” (SWW)). Unlike in English, there is no phonological reduction of vowels in German unstressed syllables (Delattre, 1969). Prominence perception on the sentence level, by contrast, is more ambiguous, as the pitch accent can be placed on every word in the phrase, depending on the communicative function of the utterance (i.e. focus or information structure). Hence, we assume the high-pitch bias to surface more strongly in perceived sentence-level prominence where top-down knowledge cannot be used to unambiguously identify the stressed word (different from word stress judgements). We will now turn to our second and most important aim of our study, i.e. the role of experience in prominence perception.

1.2. The role of experience in speech processing

As mentioned earlier, one potential source of the high-pitch bias might be the frequency of occurrence of high-pitched stressed syllables in spoken language in German (experience-based account). Frequency effects have been documented in various domains, such as language acquisition, language use, and language change (Ambridge et al., 2015; Diessel, 2007, for reviews). They have been observed on all levels of processing, ranging from the level of single sounds to larger domains (Behrens & Pfänder, 2016, p. 1). Beyond frequency, another possible mechanism we see for the high-pitch bias is the inherent psychoacoustic salience of high pitch (e.g. Evans, 2015; Honorof & Whalen, 2005; Hsu et al., 2015). Moreover, Kakouros and Räsänen (2016) argued that human stress perception is strongly guided by the temporal unpredictability of acoustic features. That is, a word is perceived as more prominent, the less predictive the f_0 and energy cues are. All of these mechanisms might be at play in prominence perception. In this paper, we set out to test the first of the outlined mechanisms in stress perception, i.e. the role of experience with high- vs. low-pitched stressed syllables (both at the word and the sentence level). To this end, we employed an exposure-test paradigm to manipulate the short-term experience with high- vs. low-pitched stressed syllables.

Exposure phases have been used successfully for speech perception research in so-called adaptation and perceptual learning studies (e.g. Eisner & McQueen, 2006; Grohe & Weber, 2016a; Hanulíková et al., 2012; Kraljic & Samuel, 2006; Norris et al., 2003;

Reinisch & Weber, 2012; Witteman et al., 2015; Witteman et al., 2014). These studies suggest that listeners quickly adapt to various kinds of properties in the input: to segments realized with regional and foreign accents (e.g. Clarke & Garrett, 2004; Evans & Iverson, 2004; Grohe & Weber, 2016a; Grohe & Weber, 2016b; Witteman et al., 2015), to different speaking styles (e.g. Poellmann et al., 2014), or to acoustic cues to stress in non-native speech (Reinisch & Weber, 2012). Most often, they tested adaptation to the properties of a single speaker. It is an open question in how far such adaptation effects also generalise to a different speaker or multiple different speakers. Some recent studies have shown that adaptation from one speaker to a single different speaker is possible to some extent (Reinisch & Holt, 2014; Xie & Myers, 2017; Xie et al., 2018b), while Bradlow and Bent (2008), for instance, demonstrate that cross-speaker generalisation to foreign accented speech is only successful for exposure with multiple speakers but not for one different single speaker. Hence, using both the same and multiple different speakers in a pre-experimental exposure phase allows us to test whether the (potential) effect of short-term exposure to high- vs. low-pitched stressed syllables is speaker-specific (and hence only occurs when the exposure phase is spoken by the same speaker as in test, henceforth “adaptation”) or whether it is speaker-independent (and hence occurs also when the exposure phase is spoken by multiple different speakers as in test, henceforth “frequency of occurrence”).

1.3. The present study

In the present study, we probe participants’ ability to identify the position of word stress (Experiment 1) and sentence stress (Experiment 2) in speech samples that are presented with different intonation contours. Word stress is lexically determined in German (i.e. there is one objectively correct answer), while sentence stress depends on prosodic prominence (the association of pitch accent with one word, its duration and intensity). We focus on whether and how experience with high vs. low-pitched stressed syllables in the input immediately before the test phase (via “exposure phase”) influences the high-pitch bias in stress judgements. The high-pitch bias is operationalised as more correct responses in the medial-peak condition, in which high pitch and stress coincide, compared to an early- or late-peak condition, in which high pitch precedes or follows the stressed syllable, respectively. We use exposure phases recorded by several different speakers than the speaker in the test phase to probe whether a potential effect of high- and low-pitched experience generalises

(cf. Pierrehumbert, 2001). A general frequency effect could affect language acquisition (both in first- and second-language contexts) and potentially also catalyse language change (for an overview see Diessel, 2007 and references therein; cf. Harrington et al., 2008). Summarising the predictions outlined in 1.1 and 1.2, we test the following hypothesis:

- i Intonation affects the identification of both word and sentence stress, such that high-pitched syllables/words are more likely to be judged as stressed.
- ii Stress judgements on the sentence level will be more strongly affected by intonation than stress judgements on the word level.
- iii The effect of intonation on stress judgements is modulated by short-term experience with high- vs. low-pitched stressed syllables, such that high-pitch exposure reinforces the high-pitch bias on both levels.

The remainder of the paper is structured as follows: Sections 2 and 3 report the results of the experiment on the perception of word stress (Experiment 1) and sentence stress (Experiment 2), respectively.³ Section 4 discusses the implications of these findings, with respect to language acquisition and language change.

2. Experiment 1: perceived position of word stress

We use a metalinguistic stress-judgment task to test the effect of experience with high- vs. low-pitched stressed syllables for the high-pitch bias, i.e. participants' ability to identify the position of word stress in different intonation conditions. We predict that listeners are generally better at indicating the stressed syllable when it is high-pitched (medial-peak contour) than when it is low-pitched (early- and late-peak contours) and that this effect of intonation condition is stronger after immediate exposure to high-pitched than to low-pitched stressed syllables. A speaker-specific adaptation account would further predict that the effect of experience is limited to (or stronger for) the same-speaker compared to the exposure to various speakers, similar to exposure effects for segmental perceptual learning (Eisner & McQueen, 2006; McQueen et al., 2006) and the intonational marking of information structure (Roettger & Rimland, 2020).

To put these hypotheses to test, we employ an exposure-test paradigm. For the test phase, we use a word stress judgement task. The exposure phases are produced by the same speaker as the stimuli in the

subsequent stress-identification task or by a number of different speakers to study effects of speaker-specific adaption vs. generalisation, see Figure 1A for an overview of the design.

2.1 Methods

2.1.1 Participants

Seventy-two German adults participated, 18 in the low-exposure same speaker condition (mean age = 24.0 years, SD = 4.1, 7 female, 11 male), 18 in the low-exposure different speaker condition (mean age = 23.2 years, SD = 3.0, 16 female, 2 male), 18 in the high-exposure same speaker condition (mean age = 23.3 years, SD = 3.8, 12 female, 6 male), and 18 in the high-exposure different speaker condition (mean age = 22.7, SD = 2.9, 12 female, 6 male).⁴ All participants had normal hearing and no prior training in intonation. Participants reported to come from different parts of Germany, predominantly from Baden-Wuerttemberg.

2.1.2 Materials and procedure

Exposure phases. The utterances for the four exposure phases were selected such that they were pragmatically suitable to be realized either with solely high-pitched accented syllables (L + H*-accents) or solely low-pitched accented syllables (L*+H-, L*-, or H + L*-accents, see examples in Figure 1C).

Two of the four exposure phases were recorded by the same female speaker as the test stimuli, two by four further female and two male speakers (20-28 years). The exposure sentences were grouped into coherent units of two to three phrases, which in turn, were grouped as a block (totalling in five coherent units per block and six blocks overall). Exposure sentences were concatenated with inter-stimulus-intervals (ISIs) of 400 ms within and 1000 ms between coherent units. Note that the high- vs. low-pitched exposure phases were judged as equally natural.⁵

The exposure phase was set up as a rating study to engage participants in the task: Participants were seated in front of a computer screen and rated the voice they heard in the previous block (~30 s per block) on various character traits (e.g. "intelligent", see Figure 1B). The exposure phase lasted three minutes and experimenters did not talk to participants between exposure and test. Exposure condition was randomly assigned to participants.

Test phase. The materials and recordings for the stress-judgement task were adapted from Experiment 1 in Zahner et al. (2019): Thirty-six trisyllabic

monomorphemic German nouns with a WSW stress pattern were used as experimental items (see Appendix A for a full list of experimental items). Additional 72 trisyllabic monomorphemic words functioned as distractors, half with a SWW and half with a WWS stress pattern. All syllables in experimental and distractor words contained full vowels to ensure that stress could potentially fall on any of the three syllables (Féry, 1998; Wiese, 1996). This choice made stress judgments more difficult as spectral vowel reduction on unstressed syllables could not be used as an (additional) cue.

The stimuli were recorded by a female speaker of Standard German (born and raised in Bremen). Afterwards we used the PSOLA-resynthesis capability (Pitch Synchronous Overlap Add, Moulines & Verhelst, 1995) of Praat (Boersma & Weenink, 2016) to manipulate the f_0 contour of each word in order to obtain medial-peak contours (resynthesized half from early-peak and half from late-peak recordings), early- and late-peak contours (resynthesized from medial-peak recordings), see Figure 1C. This was done to isolate the effects of f_0 alignment and avoid concomitant changes in duration and intensity (Kohler, 1991; Niebuhr, 2007).⁶

Figure 2 shows one experimental word in the three intonation conditions. The distractor words (36 SWW and 36 WWS words) were also resynthesized to have a comparable degree of manipulation as the experimental items and the same pitch excursion.

Experimental lists consisted of six training trials (non-words), followed by 108 stimuli, (36 experimental words and 72 distractor words). Twelve of the 36 experimental words were presented with a medial-peak, twelve with an early-peak and twelve with a late-peak contour. Intonation condition was distributed in a Latin-Square design. All 72 distractors were added to the lists to have an equal number of WSW, SWW, and WWS words.

Stimulus words were presented via headphones at a comfortable fixed loudness. The experiment was controlled using Presentation® software (Neurobehavioral Systems, Inc., Berkeley, CA, www.neurobs.com) and lasted approximately ten minutes. In each trial, a white centred fixation cross on a black screen (250 ms) was followed by a blank black screen (500 ms) with subsequent auditory stimulus presentation (see Figure 1B). The ISI between trials was 500 ms; there was no feedback and no time-out.

In the stress-judgement task, participants identified the position of the stressed syllable on a three-keyed button box. To ensure that participants understood the concept of “word stress” (German: *Betonung*), the

written instructions (see Appendix B) showed three orthographic examples in which the stressed syllable was capitalised: one SWW word (*FURie* “fury”), one WSW word (*KaNOne* “canon”), and one WWS word (*DiaMANT* “diamond”). Participants were asked to identify the stressed syllable as correctly and quickly as possible. Overall, the experiment lasted about 15 min and participants were reimbursed with a small payment or course credit.

2.2 Results

In total, we collected 2592 responses for the experimental WSW trials (72 participants x 36 experimental trials), thus exceeding the 1600 measurements recommendation by Brysbaert and Stevens (2018) or having abundant statistical power. Responses were coded as correct (button 2) or incorrect (buttons 1 or 3). Figure 3 shows the average correctness rates across exposure phases and intonation conditions. Note that a one-sample t-test with aggregated correctness scores across participants showed that subjects performed significantly above chance in all conditions (chance level = 33%, all $p < 0.0001$).

We modelled correctness using a logistic regression model (glmer) with *exposure pitch* (high vs. low), *exposure speaker* (same vs. different) and *intonation condition* (early, medial, late) as predictors using the *lme4* package (Bates et al., 2005) in *R* (version 3.6.2; R Development Core Team, 2015). *Participants* and *items* were entered as crossed random effects (Baayen et al., 2008). The initial model included main effects and interaction terms of the predictors and random intercepts for *participants* and *items* (Bates et al., 2015; Matuschek et al., 2017). Non-significant predictors and interactions were removed, and the model was refitted. Where possible, random slopes were added to the model if they improved the model fit (in terms of LogLikelihood as indicated in model comparisons with the *anova()* function). In case the model did not converge with random intercepts for *participants* and *items*, the random intercept for *items* was removed. The model structure of the final models is reported in the analysis script on Mendeley (<https://doi.org/10.17632/gjcxmdpsbj.1>).

The final model showed a significant interaction between *exposure pitch* and *intonation condition* ($\chi^2(2) = 7.3, p = 0.027$) and a main effect of *intonation condition* ($\chi^2(2) = 56.3, p < 0.0001$). There was no effect of *exposure speaker* ($p = 0.62$) and no interaction between *exposure speaker* and *intonation condition* ($p = 0.15$), no effect of *exposure pitch* ($p = 0.46$) and no

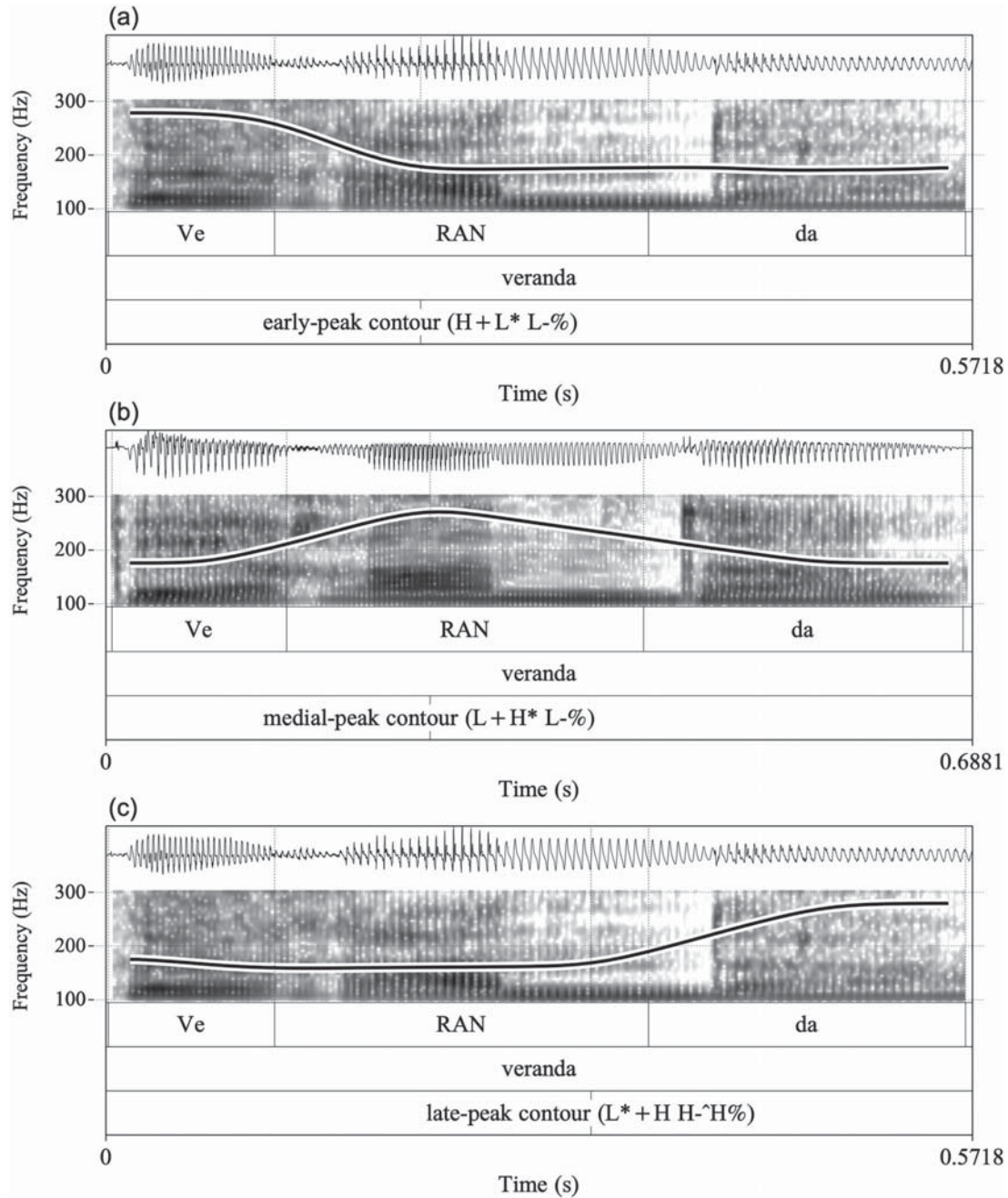


Figure 2. Sound pressure wave, spectrogram and PSOLA-resynthesized intonation contours (smoothed with bandwidth 10) for one experimental word (capitals indicate lexical stress; in (a) early-peak condition, (b) medial-peak condition and (c) late-peak condition).

interaction between *exposure pitch* and *exposure speaker* ($p = 0.92$).

To further investigate the interaction between *exposure pitch* and *intonation condition*, we split the data by *exposure pitch*: irrespective of *exposure pitch*, words in the early- and late-peak conditions led to fewer correct responses than in the medial-peak condition. Importantly, the effect of *intonation condition* was stronger after high-pitch exposure (medial vs. early: $\beta = -1.25$, $SE = 0.17$, $z = -7.4$, $p < 0.0001$; medial

vs. late: $\beta = -0.88$, $SE = 0.17$, $z = -5.3$, $p < 0.0001$) than after low-pitch exposure (medial vs. early: $\beta = -0.62$, $SE = 0.16$, $z = -3.9$, $p < 0.0001$; medial vs. late: $\beta = -0.46$, $SE = 0.16$, $z = -2.9$, $p = 0.003$).

Note that the observed error pattern in the erroneous responses is compatible with the high-pitch bias: Participants more often judged the first syllable as stressed in the early-peak condition and the third syllable as stressed in the late-peak condition, see Figure 4.

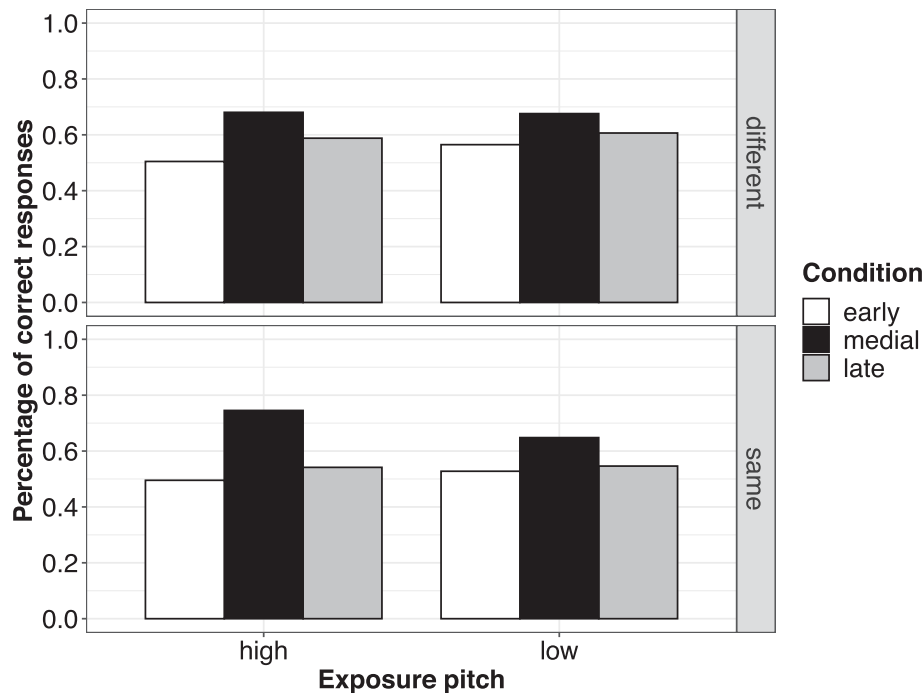


Figure 3. Average correctness rates across intonation (early-, medial- and late-peak) and exposure conditions (high vs. low exposure spoken by the same vs. various speakers as test stimuli) for word stress.

2.4 Discussion

We first discuss general aspects of the results and then turn to the novel part, the role of experience. As expected, since it replicates earlier findings, Experiment 1 showed that listeners make more erroneous stress judgments when the f_0 peak is not realized on the stressed syllable and that errors are biased towards the syllable with the

peak (cf. Zahner et al., 2019, Experiment 1). Note that in all three intonation conditions (early-, medial- and late-peak), correctness in participants' stress judgments was above chance. However, it was never at ceiling level, not even in the "ideal" condition, in which the stressed syllable was high-pitched (i.e. medial-peak condition). This might be due to the fact that we used words with a

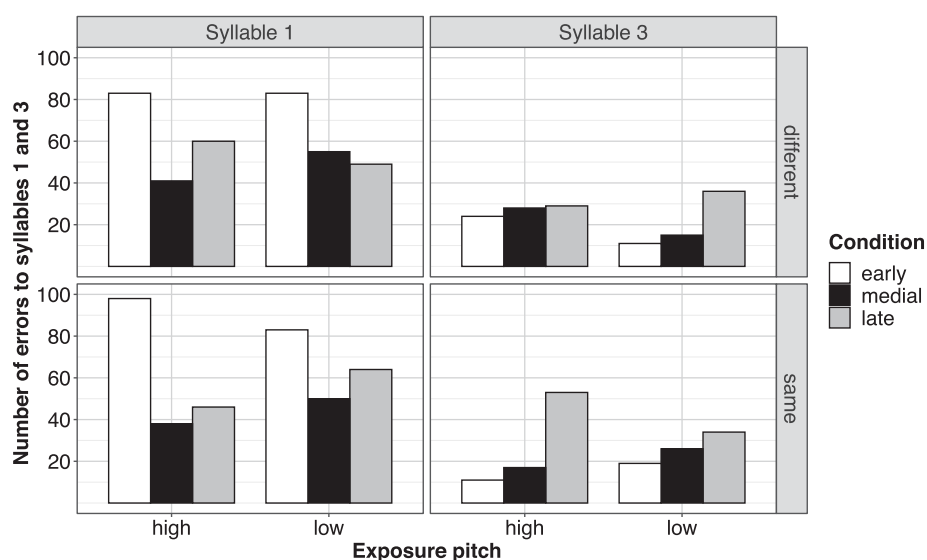


Figure 4. Number of erroneous syllable-1- and syllable 3-responses across intonation and exposure conditions for perceived word stress (Experiment 1).

relatively low frequency of occurrence, diminishing the availability of top-down information based on lexical knowledge. Overall, these findings can neither be explained by the usage of exclusively top-down information, nor by an exclusively signal-driven approach.

More importantly, we were able to unravel one possible source for this high-pitch bias in the perception of word stress: the frequency of occurrence of pitch accents in the immediate input. The correctness of stress judgements was influenced by whether stressed syllables were high- or low-pitched in the exposure phase, but not by *who* produced the tokens (same or different speakers). This suggests that listeners' responses are not caused by speaker-specific perceptual retuning or adaptation to the speakers' way of producing stressed syllables (cf. Bradlow et al., 1999; Xie & Myers, 2017), but listeners generalised the association (or dissociation) between high f0 and stress to other speakers of the language (cf. Bradlow & Bent, 2008; Xie et al., 2018a; Xie et al., 2018b). While the influence of exposure to different pitch accent types on sentence comprehension is speaker-sensitive and hence restricted to the speaker a listener is currently exposed to (Roettger & Rimland, 2020), the exposure effect we find for correctness rates in stress judgments generalises. It has thus more far-reaching implications, not only for cross-linguistic speech processing, but also for language change and developmental aspects. We will come back to these aspects in the General Discussion (Section 4). The remaining question for now is whether our findings extend to stress perception on the level of the phrase.

3. Experiment 2: perceived position of sentence stress

In Experiment 2, we investigated the role of exposure for the use of high pitch in identifying sentence stress, i.e. the most prominent word within the utterance. To ensure comparability with Experiment 1 in terms of materials, we used *wh*-questions consisting of three monosyllabic words. *Wh*-questions can have the *wh*-word as the stressed word, or the verb, or the final word, see example (1). In (1)a, the question focuses the agent, e.g. in a situation in which the agent is unknown or in an echo-question context, in which the speaker expresses disbelief that this particular agent performed the action. (1)b is compatible with a situation in which the lexical semantics of the verb is in question ("paying" as opposed to "claiming" money) or its polarity ("paying" as opposed to "not paying"). (1)c is expected in a condition in which the identity of the object is questioned. Note that *wh*-questions in German can end in a

final rise or a final fall (Braun et al., 2019; Kohler, 2004), which allows us to use the same intonation conditions as in Experiment 1.

- | | | | | |
|-----|-------|------------------------|----------------------|------------------------|
| (1) | a. | [Wann] _F | zahlt | Chris? |
| | b. | Wann | [zahlt] _F | Chris? |
| | c. | Wann | zahlt | [Chris] _F ? |
| | Engl. | "When does Chris pay?" | | |

We used the same high- and low-exposure phases as in Experiment 1. Since speaker identity did not influence participants' responses in Experiment 1, we only employed the different-speaker exposure phases.

3.1 Methods

3.1.1 Materials

We selected a set of 54 transitive and 54 intransitive monosyllabic German verbs from the CELEX lexical database (Baayen et al., 1993). The sentences containing transitive verbs ended in an object noun or proper name, those with intransitive verbs in an adverbial. We constructed 108 *wh*-questions consisting of three monosyllabic words (e.g. "Who loves Sue?"). To increase variability, 32 questions had the *wh*-pronoun *wer* ("who"), 18 *wo* ("where"), 23 *was* ("what"), 15 *wann* ("when") and 12 *wie* ("how"). Of these 108 *wh*-questions, we used 36 as experimental items (with a pitch accent on the verb, conceptually identical to the WSW experimental trials in Experiment 1, see Appendix C), and 72 as distractors (36 with an accent on the question pronoun, similar to the SWW words in Experiment 1, and 36 with an accent on the final word, similar to the WWS words in Experiment 1). The type of question pronoun and verb type (transitive or intransitive) was matched across accent positions. The 36 experimental trials were checked for whether an accent on the verb sounds natural and items were changed if necessary (this was done by an independent researcher not involved in this project). We constructed three additional *wh*-questions to be used as practice trials.

Recording. All 108 *wh*-questions were recorded by a trained male speaker of Standard German.⁷ He read the 36 experimental items with an accent on the verb three times, with three different accent types, one high-pitched (L + H*) and two low-pitched accents (H + L* and L* + H). In the same recording session, he read the 36 items with an accent on the *wh*-word in two conditions: L + H* (medial-peak) and L* + H (late-peak) and the 36 items with an accent on the final verb in two conditions: L + H* (medial-peak) and H + L* (early-peak). The practice items were all recorded with an L + H* accent.

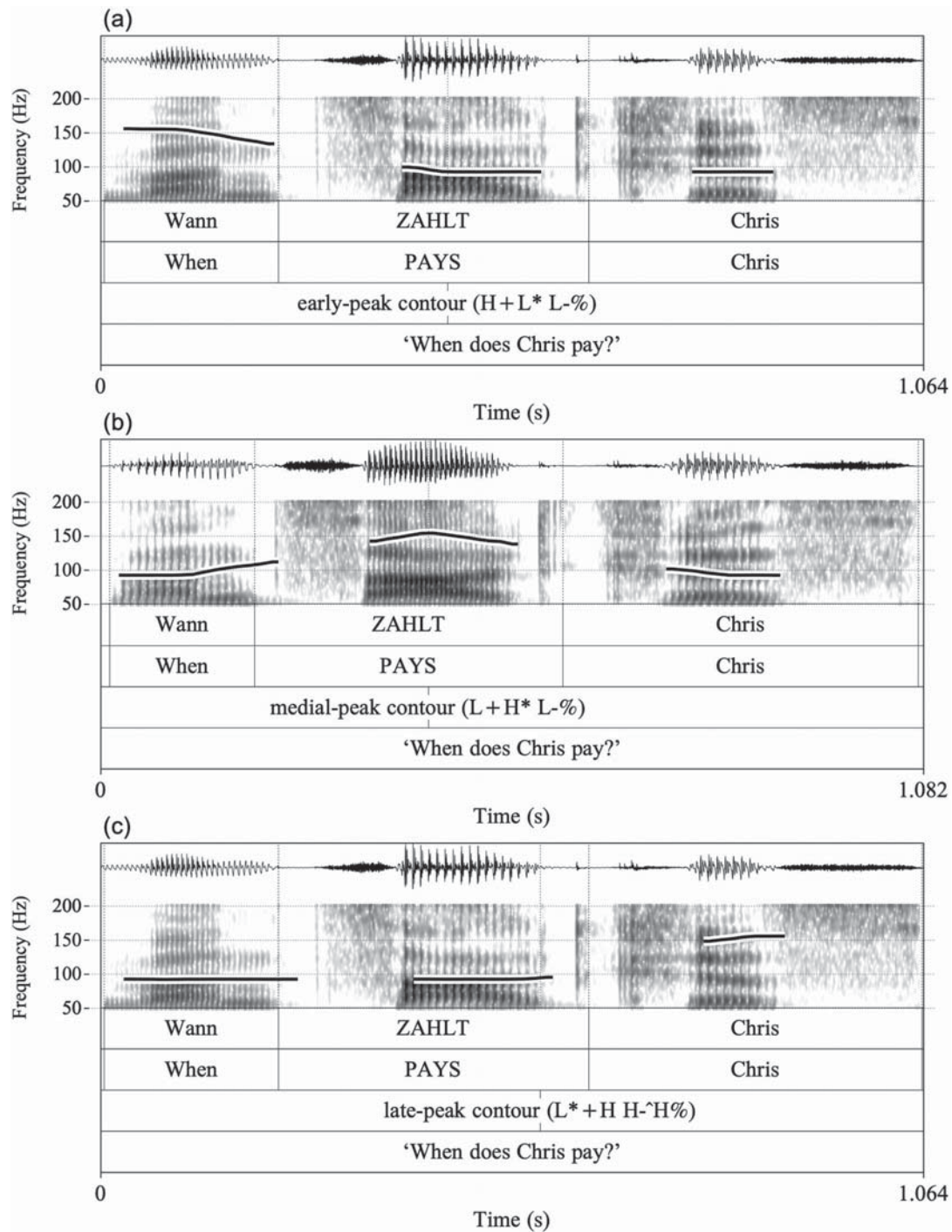


Figure 5. Sound pressure wave, spectrogram and PSOLA-resynthesized intonation contours (smoothed with bandwidth 10) for one experimental item (capitals indicate the stressed word in (a) early-peak condition, (b) medial-peak condition and (c) late-peak condition).

Manipulation. As in Experiment 1, the recordings were annotated at the syllable level and the vowels were marked. The f_0 -contour was annotated for minima and maxima. The manipulation was done in the same manner as in Experiment 1. Figure 5 shows one experimental *wh*-question (with stress on the verb) in the three intonation conditions.

3.1.2 Procedure

We used the same procedure as in Experiment 1 to compile the experimental lists and the exposure phase was also the same as in Experiment 1. In the test phase, participants were instructed to identify the stressed word (German: *betontes Wort*) within the sentence (see Appendix B for the exact instructions

given to participants). The experiment was programmed and run using FindingFive (<https://eu.findingfive.com>). Half of the participants in each exposure condition were recruited via the MTurk linkage of FindingFive, half from the subject pool at the University of Konstanz (via a FindingFive account). The experiment lasted on average 11 min and participants were reimbursed with a small payment or course credit.

3.1.3 Participants

Forty-three participants took part in the experiment. We excluded five participants whose native language was not German or who were bilingual and two participants who were older than 40 years. This left 36 participants, 18 in the low-exposure condition (mean age = 25.4 years, SD = 4.7 years, 7 female, 11 male), 18 in the high exposure condition (mean age = 26.8 years, SD = 5.4 years, 7 female, 10 male, 1 diverse). As in Experiment 1, participants had normal hearing, no prior training in intonation and reported to come from different parts of Germany, predominantly from Baden-Wuerttemberg.

3.2 Results

In total, we collected 1296 responses for the experimental trials with sentence stress on the verb (36 participants $\times$ 36 experimental trials). Figure 6 shows the average correctness rates across exposure phases (exposure to high- vs. low-pitched stressed syllables) and intonation conditions (early-, medial- and late-peak condition) for sentence stress judgements.

A one-sampled t-test averaged over participants indicated that participants performed above chance (chance

level = 33%) in the medial-peak condition only ($p < 0.0001$); for the early- and late-peak conditions participants performed significantly below chance (all $p < 0.0001$). Statistical analysis of correctness was identical to Experiment 1. The final model showed a significant interaction between *exposure pitch* and *intonation condition* ($\chi^2(2) = 9.0, p = 0.011$) and a main effect of *intonation condition* ($\chi^2(2) = 185.7, p < 0.0001$).

To further investigate the interaction, we split the data by *exposure pitch*: irrespective of *exposure pitch*, words in the early- and late-peak conditions led to fewer correct responses than in the medial-peak condition. Importantly, the effect of *intonation condition* was stronger after high-pitch exposure (medial vs. early: $\beta = -3.06, SE = 0.26, z = -11.6, p < 0.0001$; medial vs. late: $\beta = -3.23, SE = 0.27, z = -11.9, p < 0.0001$) than after low-pitch exposure (medial vs. early: $\beta = -2.24, SE = 0.23, z = -9.7, p < 0.0001$; medial vs. late: $\beta = -2.08, SE = 0.22, z = -9.3, p < 0.0001$), see Figure 6. There was no effect of *verb type* (transitive vs. intransitive), $p = 0.07$. The effect of *wh*-type could not be tested due to convergence issues.

The effect of *intonation condition* seems larger overall in Experiment 2 than in Experiment 1 (cf. Figures 3 and 6). To quantify this difference, we combined the data from the different-speaker exposure and the current data set and tested for interactions between the predictors *stress type* (word or sentence stress), *exposure pitch* (high vs. low) and *intonation condition* (early-, medial- or late-peak). There was a significant interaction between *stress type* and *intonation condition*, with participants being more strongly affected by *intonation condition* when judging sentence stress (Experiment 2) as compared to word stress (Experiment 1, $\chi^2(2) = 80.6, p < 0.0001$). *Exposure pitch* did not influence the effect of *intonation condition* differently in the two experiments (three-way-interaction $p = 0.15$). Overall, participants showed higher correctness in Experiment 1 than in Experiment 2 ($\chi^2(1) = 7.5, p = 0.006$).

The error patterns for Experiment 2 are similar to what we observed in Experiment 1: Participants more often indicated a high-pitched word as the stressed one, see Figure 7.

3.3 Discussion

So far, research had mainly focused on the degree of prominence that is expressed by different accent types (Baumann & Winter, 2018), as an exception see Kutschheid et al. (2021): Similar to the effect of intonation condition in word stress judgements, pitch accent type also influenced which word on the sentence level participants perceived as most prominent. This finding is of

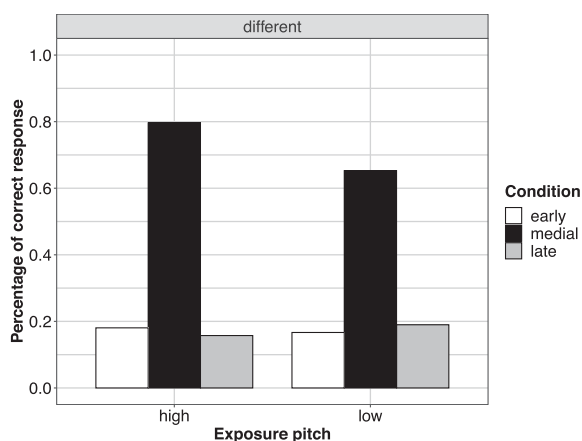


Figure 6. Average correctness rates across intonation (early-, medial- and late-peak) and exposure conditions (high vs. low exposure) for sentence stress. Note that only different speaker exposure conditions were used in Experiment 2.

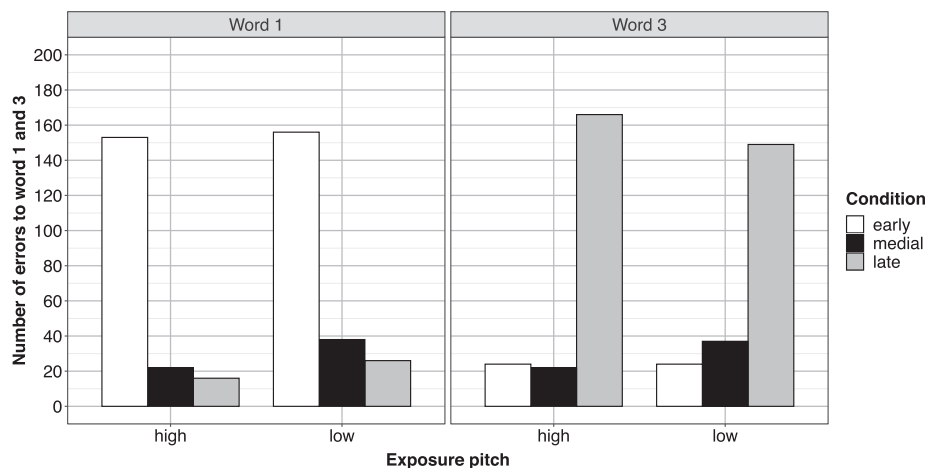


Figure 7. Number of erroneous word-1 – and word 3-responses across intonation (early-, medial- or late-peak) and exposure conditions (high- vs. low-pitch) for perceived sentence stress (Experiment 1). Note that only different speaker exposure conditions were used in this experiment.

direct importance for speech comprehension, since perceiving the sentence stress on a non-intended word may change the inferred information structure (as focus is expressed by pitch accents) and may hence hamper successful communication. In general, correctness was lower at the phrasal level than at the word level, which can be explained by the fact that there is no objectively correct answer because, in theory, each word can be made prominent by a pitch accent, whereas only one syllable in a word is the lexically stressed one. The use of nonce words in Experiment 1⁸, as recommended by the reviewers, would presumably lead to similar correctness rates as we see in Experiment 2.

Regarding our main aim, the role of exposure, our results reveal that the high-pitch bias was clearly affected by exposure type (high- vs. low pitched stressed syllables/words). The effect of intonation condition was larger for the exposure with only high-pitched stressed syllables compared to the low-pitch exposure phase, despite the fact that the stimuli in exposure and test came from different speakers. This corroborates the finding that the frequency of occurrence of accent types in the immediate input affects the identification of the stress position. In other words, occurrence frequency seems to be one potential source of the high-pitch bias.

4. General Discussion

In the Introduction we identified two open questions to be answered in this paper: whether stress perception behaves similarly on the word and phrasal level and which role exposure to different pitch accent types plays in modulating the effect of intonation on stress

perception. Regarding the first question, the current study shows that listeners more readily identify high-pitched syllables as stressed, both at the word level (Experiment 1) and at the sentence level (Experiment 2). In line with our expectations, in the word context, listeners frequently mis-identified the position of lexical stress (which was realised on the second syllable in trisyllabic words) on the first syllable in early-peak accents (H + L*) and on the last syllable in late-peak accents (L*+H). Likewise, in the utterance context (“Who loves Sue?”), listeners often mis-identified the position of the stressed word (i.e. the verb) on the *wh*-word in early-peak accents (H + L*) and on the final word in late-peak accents (L*+H). Previous research on prominence perception has shown an intricate interplay between bottom-up acoustic cues and top-down contextual cues (Bishop et al., 2020; Cole et al., 2010; Wagner, 2005). We have only tested the former effect, but it would be interesting to test how exposure affects the high-pitch bias in supportive and non-supportive contexts. We leave this aspect to future research.

With regard to the second question, the main goal of this study was to investigate how immediate exposure affects the effect of intonational realisation for stress detection (both on the word and sentence level). We showed that listeners who were exposed to a three-minute stretch of utterances with solely high-pitched stressed syllables relied more on high pitch as a stress cue than listeners who were exposed to the same stimuli with low-pitched stressed syllables. An alternative approach to test this hypothesis would have been to compare stress processing in listeners from languages with different frequency distributions of pitch accent types. Frequency estimates of accent types in a language

could be retrieved from annotated speech corpora: In an (American) English radio news corpus, around 90% of the accented syllables are high-pitched (Dainora, 2006), i.e. listeners receive substantial evidence for the association of high pitch and stress. In German appointment-making dialogues, high-pitched accents sum up to approximately 60% (Peters et al., 2005, p. 12). However, since these corpora are not comparable in terms of situational context and sentence types (e.g. monologue vs. dialogue, information exchange vs. social interaction), it is difficult to use these frequency estimates as a proxy for the occurrence frequency in the input. To circumvent this difficulty, we decided to use an exposure-test paradigm to manipulate the short-term exposure to high-pitched vs. low-pitched stressed syllables.

Still, the described frequency effect of intonation may have repercussions for how native speakers of languages with foremost low-pitched stressed syllables, e.g. Bern and Valais Swiss German or Glaswegian English (Fitzpatrick-Cole, 1999; Leemann, 2012; Leemann & Siebenhaar, 2007; Smith & Rathcke, 2020), perceive lexical and phrasal prominence, compared to languages or varieties with foremost high-pitched stressed syllables (e.g. Standard German, American English). Extrapolating from our laboratory findings, we predict a smaller effect of intonation condition on stress judgements for languages or varieties with a high proportion of low-pitched stressed syllables (Kutscheid et al. (2019) for preliminary data on Swiss German).

The effect of exposure was larger for sentence stress than for word stress. A possible explanation for this is that there is an unambiguously correct answer for lexical stress, while all accent positions are possible for sentence stress. In future studies, it would be useful to replicate the distributional effect of intonation with different languages, some that have a high and a low proportion of high-pitched stressed syllables, e.g. Standard German vs. Swiss German (Fitzpatrick-Cole, 1999; Leemann, 2012; Peters et al., 2005). Given the differences in effect size, it would be vital to show this proof-of-concept with tasks that tap into the identification of phrasal prominence.

Importantly, we were able to show that the effect of exposure (high vs. low-pitched stressed syllables) on perceived word and sentence stress generalises to different speakers. Hence, differently from what has recently been shown on the adaptive processing of intonational variation for sentence comprehension (Roettger & Rimland, 2020), the adjustment of cue weights for the processing of word and sentence stress seems to be speaker-independent. Based on what is known on the adaptation to foreign accents (Bradlow & Bent, 2008), the usage of

multiple different speakers for our different speaker exposure condition, instead of one single different speaker, might have been particularly beneficial for listeners. In any case, our findings speak in favour of a general frequency effect – a finding that has implications for language acquisition and language change, which we will briefly address in the following.

As for language acquisition, children who are exposed to predominantly high-pitched stressed syllables may quickly rely on high f_0 as a cue to word stress (Quam & Swingle, 2014; Zahner et al., 2016) and may find it harder to use stress in other contexts than children who are exposed to a wider variety of accent types. In this scenario, the utility of stress for speech processing may decrease over time. For L2 acquisition, listeners from languages with a large proportion of high-pitched stressed syllables may misperceive the prominence structure (and consequently sentence accent) in an L2 that has a considerable amount of low-pitched stressed syllables, which might lead to comprehension difficulties. For instance, Tone 4 in Mandarin (high falling) is often interpreted as assertive and hence presumably as prominent by German and English listeners (Braun & Johnson, 2011; Broselow et al., 1987).

Finally, misperceived sentence stress, possibly leading to misperceived focus structure, may eventually also trigger language change such that accent placement regularities might be blurred. Ohala (1981, 1993), for example, suggested that listeners play a central role in sound change. Listeners may misattribute phonetic features in the acoustic signal to a segment that was originally not intended by the speaker. In our case, listeners might interpret the feature high pitch (rather than the word that the pitch accent is associated with, i.e. the stressed word) as prosodically prominent and thus possibly misinterpret it as focus. As a potential case in point, many native speakers of English often accentuate prepositions, e.g. “I’ll present this TO Sam” – which from an information-structural perspective is odd. Diachronically, this tendency may have emerged from a misinterpretation of the accented words as medial peaks instead of early peaks and finally transferred into more recent versions of English. Admittedly daring, such hypotheses might inspire diachronic analyses.

To conclude, we present a study that identifies high pitch and experience with high-pitched stressed syllables as a main mechanism that guide stress perception at the word and sentence level. In future research, we plan to test how long-lasting the effect of exposure is, whether listeners from languages with different accent distributions differ in their use of f_0 for stress perception and whether or not trained annotators of intonation are more immune to immediate exposure than naïve

listeners. A second line of research concerns the role of experience on stress perception in online processing (e.g. visual world eye-tracking or EEG measurements), which will allow us to tap into the actual representations and processing mechanisms.

Notes

1. Note that syllables can carry primary or secondary stress and listeners make use of both in speech perception (Mattys, 2000).
2. A number of production studies successfully disentangled phrase-level accent and word stress (see e.g., Gordon, 2014; Sluijter & van Heuven, 1996; Sluijter et al., 1997) and identified the cues to stress in the presence and absence of a sentence accent. In these studies, high pitch turned out to play a minor role for lexical stress perception in unaccented position. Despite the empirical evidence, listeners are biased towards interpreting high-pitched syllables as stressed (Friedrich et al., 2001; Schwab & Dellwo, 2017; Zahner et al., 2019).
3. The study is approved by the IRB of the University of Konstanz (IRB 30/2016); informed written consent was obtained from all participants.
4. A power test for 18 randomly selected participants of Experiment 1b in Zahner et al. (2019) – a similar button-box experiment with three intonation conditions – shows that 18 participants are sufficient to achieve an average power of 94.0% (95%CI [83.5%; 98.8%]); the CI is based on 50 simulations of the final model for Experiment 1b using the R-package *simr* (Green & MacLeod, 2016).
5. Eight participants judged the naturalness of 24 randomly selected semantic units (12 from the high- and 12 from the low-exposure condition) on a scale from 1 (= very unnatural) to 5 (=very natural). A linear mixed effect regression model indicated no effect of pitch ($p = 0.33$; on average high-exposure sentences received a score of 4.1, $SD = 1.0$, low-exposure sentences 4.7, $SD = 0.7$).
6. Specifically, Niebuhr (2007) showed that in natural productions, syllables on which the f_0 peak is realized may additionally be produced with higher intensity and longer duration – cues that also signal stress. Thus, the acoustic difference between stressed and unstressed syllables tends to be reduced.
7. The change in speaker was necessary because the speaker of Experiment 1 was no longer available.
8. The design in Experiment 1 (real low-frequency words with only one correct answer for the stressed syllable) is a stronger test case for the high-pitch bias since top-down knowledge needs to be overruled for the effect to surface – which is indeed what we show in Experiment 1.

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